Shashank S

AI / Machine Learning Engineer | Gen AI Engineer

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Experience

Digitomics Nov 2025 – Present

Founding Team Member — AI Engineer *Bengaluru, India*

- Architecting a **production-grade GenAI platform** with FinOps, DevOps, and Generic agents, implementing **intent classification, task-based model routing, tool calling, RAG, and HITL workflows**.
- Implemented **task-aware model routing** using newer low-latency models including **Jev and GPT-5.6 Luna**, reducing intent-classification latency from **multi-second inference to sub-second responses**.
- Built a **real-time voice bot using GPT Realtime**, enabling low-latency voice interactions that follow the same **intent, retrieval, and agent workflow** as the platform's text-based chatbot.
- Built a **multi-tenant RAG pipeline** with contextual chunking, multi-query retrieval, vector search, and **tenant-aware knowledge bases** for secure document retrieval.
- Developed **FinOps and DevOps agents** with database/API integrations, approval workflows, prompt optimization, and **platform-wide rate limiting**.
- Engineered the **end-to-end tenant subscription flow**, covering plan selection, onboarding, subscription state, access control, and tenant isolation.
- Deployed **GPT-5.6 Luna through Azure AI infrastructure** and managed cloud operations, including **VM maintenance, environment configuration, and Stage-to-Production deployments**.
- Contributed to **backend and AI development** using FastAPI, databases, and cloud services, taking features from experimentation to production.
- Led and mentored **interns and team members** through architecture discussions, code reviews, technical guidance, and **end-to-end feature ownership**.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL
AI/GenAI: LLMs, NLP, Transformers, PyTorch, Hugging Face, Generative AI, Prompt Engineering, Fine-Tuning
RAG/Agents: RAG, Embeddings, Vector Search, Hybrid Search, Reranking, FAISS, ChromaDB, pgvector, LangChain, LangGraph, LlamaIndex, MCP, Tool Calling, Multi-Agent Systems
LLM Evaluation: Ragas, DeepEval, LangSmith, Langfuse, LLM-as-a-Judge, Hallucination Detection, RAG Evaluation
Backend: FastAPI, Node.js, Express.js, REST APIs, GraphQL, Microservices, Redis, Kafka, gRPC, Async Processing
Databases: PostgreSQL, MongoDB, MySQL, Elasticsearch, Redis, Vector Databases
ML/MLOps: Scikit-learn, TensorFlow, MLflow, Docker, Kubernetes, CI/CD, Model Serving, Model Monitoring
Cloud: AWS, Azure, Azure OpenAI, Azure AI Foundry
Tools: Git, GitHub, Linux, PyTest, Postman, System Design

Projects

RAG-Based Document Q&A System | *LangChain, OpenAI, ChromaDB, Python, REST API* | GitHub 2025

- Built a Retrieval-Augmented Generation system using LangChain, OpenAI embeddings, and ChromaDB to retrieve relevant document context and generate grounded answers.
- Evaluated 500+ queries and achieved 92% context accuracy while reducing hallucinations by 40%; exposed the pipeline through a REST API with sub-2s response time.

AI Voice Calendar Agent | *React, TypeScript, FastAPI, Azure OpenAI Realtime, WebRTC, PostgreSQL, Docker* | GitHub 2026

- Built a multi-user voice agent that signs users in with Google OAuth, connects Calendar in the same flow, and schedules meetings, calendar blocks, and reminders through Azure OpenAI Realtime over WebRTC with barge-in.
- Grounded actions in FastAPI tools and Postgres so meeting facts and Google Meet links come only from Calendar API responses; optional candidate invites via calendar email updates, with Dockerized React/nginx frontend and FastAPI/Postgres backend.

Shuttle: Badminton Domain LLM Fine-Tuning | *Python, PyTorch, PEFT/LoRA, SmolLM2* | GitHub 2026

- Fine-tuned SmolLM2-360M-Instruct on a 110-example badminton Q&A dataset using LoRA/PEFT, training only $\sim$1.2% of parameters ($\sim$4.3M/366M) and cutting training loss from $\sim$3.0 to $\sim$2.2 over 3 epochs on CPU.
- Built an end-to-end pipeline (dataset $\rightarrow$ LoRA training $\rightarrow$ inference) and shipped a FastAPI-backed web chat UI for domain Q&A on rules, strokes, tactics, and equipment.